These approximate search methods are indeed a tradeoff between accuracy and performance; there are a few different (and sometimes complementary) strategies that can be used to have faster approximate queries:

- **Reduce the dimensionality of the objects**: using algorithms like PCA or Discrete Fourier Transform, it is possible to project the dataset's object into a different, lower-dimensional space. The idea is that this space will maintain only the essential information to distinguish between different points in the dataset. With dynamic datasets, this method is obviously less effective.
- **Reduce the number of branches traversed**: as we have seen in the previous section, our pruning strategy is quite conservative, meaning that we traverse a branch if there is any chance (even the slightest) that we can find a point in that branch closer than our current nearest neighbor. By using a more aggressive pruning strategy we can reduce the number of branches (and ultimately dataset points) touched, as long as we accept that our result could not be the most accurate possible.
- **Use an early termination strategy**: in this case, the search is stopped when the current result is judged good enough. The criterion to decide what's "good enough" can be a threshold (for instance, when a NN closer than some distance is found), or a stop condition connected to the probability of finding a better match (for instance, if branches are visited from closer to further WRT the query point, this probability decreases with the number of leaves visited).